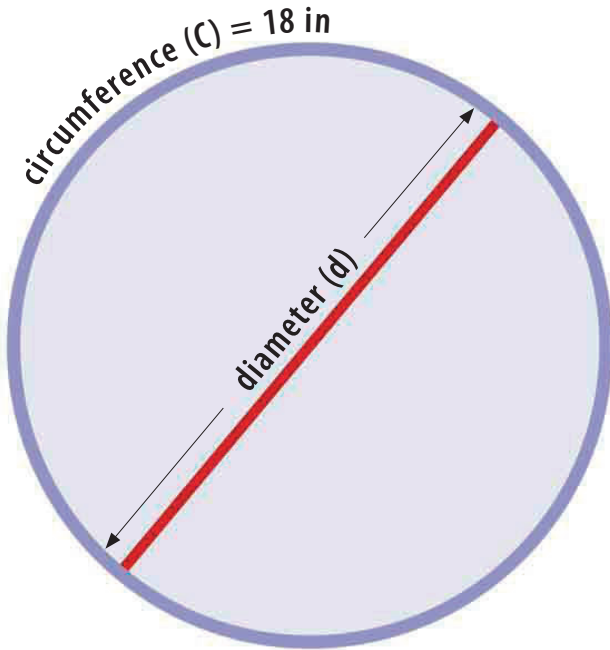


Diameter (d)

The diameter is the distance across the middle of a circle. It is twice the length of the radius. A circle's diameter can be found by doubling the length of its radius, or by using its circumference and the number pi in the formula shown below. The formula is a rearranged version of the formula for the circumference of a circle.



△ Finding the diameter

This circle has a circumference of 18 in. Its diameter can be found using the formula given above.

The formula for diameter shows that the length of the diameter is equal to the length of the circumference divided by the number pi.

$$d = \frac{C}{\pi}$$

Labels: diameter (pointing to d), circumference (pointing to C), π is a constant (pointing to π)

Substitute known values into the formula for diameter. In the example shown here, the circumference of the circle is 18 in.

$$d = \frac{18}{\pi}$$

Divide the circumference by the value of pi, 3.14, to find the length of the diameter.

$$d = \frac{18}{3.14}$$

more accurate to use π button on a calculator

Round the answer to a suitable number of decimal places. In this example, the answer is given to two decimal places.

$$d = 5.73 \text{ in}$$

the answer is given to two decimal places

LOOKING CLOSER

Why π ?

All circles are similar to one another. This means that corresponding lengths in circles, such as their diameters and circumferences, are always in proportion to each other. The number π is found by dividing the circumference of a circle by its diameter—any circle's circumference divided by its diameter always equals π —it is a constant value.

▷ Similar circles

As all circles are enlargements of each other, their diameters (d_1 , d_2) and circumferences (C_1 , C_2) are always in proportion to one another.

